

RAG based Q&A bot

ABSTRACT

This project presents a Retrieval-Augmented Generation (RAG) based Question Answering bot that provides accurate and meaningful answers from user-uploaded documents such as PDFs, notes, and reports. Unlike general AI chatbots that rely on internet knowledge and may produce irrelevant or incorrect responses, this system focuses only on the content of the documents provided by the user.

The system works by extracting text from the uploaded documents, converting the text into vector representations, and storing them in a vector database. When a user asks a question, the system retrieves the most relevant parts of the documents using semantic search and then generates an answer based strictly on the retrieved content. This approach improves accuracy and reduces the chances of incorrect or unrelated answers.

To enhance reliability and usability, the system shows the source of the answer along with page references, supports multiple documents as a single searchable knowledge base, and provides answers in different formats such as simple, detailed, or exam-oriented explanations. In addition, a confidence score is displayed to indicate how relevant the retrieved information is to the user's question.

Overall, this project offers an efficient and reliable solution for understanding and retrieving information from large documents, making it especially useful for academic study, research, and professional applications.